Executive Summary

This project analyzes sales data to identify high-performing product categories, profitable regions, customer segments, and the impact of discounts on profitability. The analysis was conducted using Python, Pandas, Matplotlib, and Seaborn.

The findings reveal that Office Supplies generated the highest sales and profit, while Furniture showed lower profitability. The Consumer segment contributed the highest revenue, and sales peaked during the final months of the year. The study also found a negative relationship between discounts and profit, indicating that excessive discounting reduces profitability.

Based on the analysis, recommendations include optimizing discount strategies, focusing on high-performing categories, strengthening customer retention, and expanding business efforts in underperforming regions.

Dataset Overview

- Dataset: Global Superstore Sales Dataset
- Total Records: 51,290
- Features: 21
- Missing Values: 0
- Tools Used: Python, Pandas, Matplotlib, Seaborn

Sales Performance Analysis – Visualizations & Insights

[1]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

pd.set_option('display.max_columns', None)

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

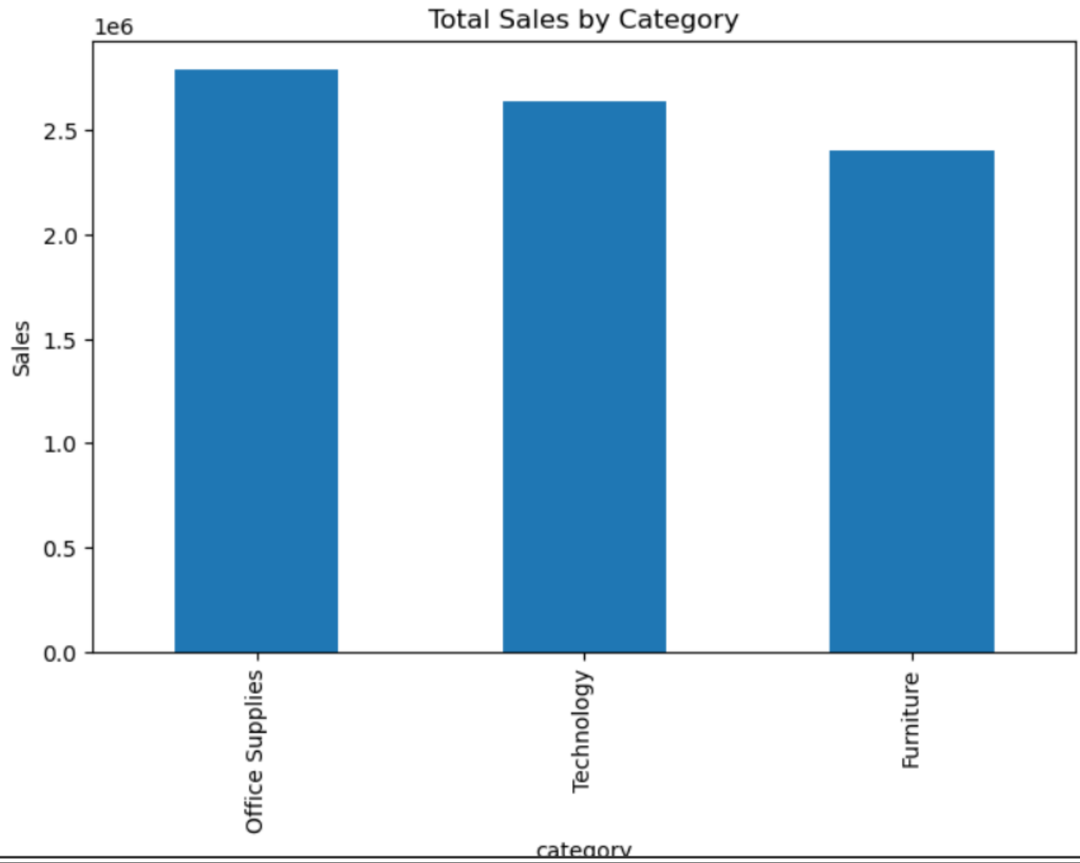
```python
df = pd.read_csv(
    r"D:\DA INTERN\da vs\Sales performance analysis\SuperStoreOrders.csv"
)

df.head()
```

	order_id	order_date	ship_date	ship_mode	customer_name	segment	state	country	market	region	product_id	category	sub_category	product_nai
0	AG-2011-2040	1/1/2011	6/1/2011	Standard Class	Toby Braunhardt	Consumer	Constantine	Algeria	Africa	Africa	OFF-TEN-10000025	Office Supplies	Storage	Tenex Locke B
1	IN-2011-47883	1/1/2011	8/1/2011	Standard Class	Joseph Holt	Consumer	New South Wales	Australia	APAC	Oceania	OFF-SU-10000618	Office Supplies	Supplies	Acme Trimm High Spe
2	HU-2011-1220	1/1/2011	5/1/2011	Second Class	Annie Thurman	Consumer	Budapest	Hungary	EMEA	EMEA	OFF-TEN-10001585	Office Supplies	Storage	Tenex B Single Wi
3	IT-2011-3647632	1/1/2011	5/1/2011	Second Class	Eugene Moren	Home Office	Stockholm	Sweden	EU	North	OFF-PA-10001492	Office Supplies	Paper	Enermax N Car Premi
4	IN-2011-47883	1/1/2011	8/1/2011	Standard Class	Joseph Holt	Consumer	New South Wales	Australia	APAC	Oceania	FUR-FU-10003447	Furniture	Furnishings	Eldon Li Bulb, Duo P

1. Sales by Category

]:
```python
category_sales = df.groupby('category')['sales'].sum().sort_values(ascending=False)

print(category_sales)

plt.figure(figsize=(8,5))
category_sales.plot(kind='bar')

plt.title("Total Sales by Category")
plt.ylabel("Sales")
plt.show()
```

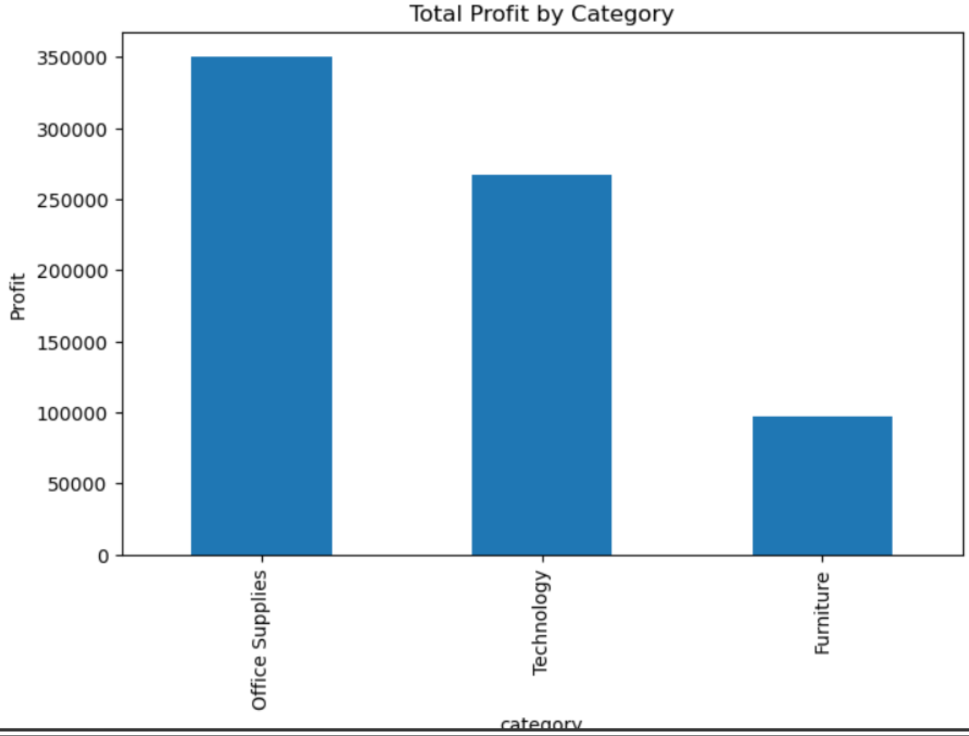
```
category
Office Supplies    2790258.0
Technology         2638265.0
Furniture          2406605.0
Name: sales, dtype: float64
```



2. Profit by Category

```
15]: category_profit = df.groupby('category')['profit'].sum().sort_values(ascending=False)

print(category_profit)

plt.figure(figsize=(8,5))
category_profit.plot(kind='bar')

plt.title("Total Profit by Category")
plt.ylabel("Profit")
plt.show()
```

```
category
Office Supplies     350107.32450
Technology          267573.47238
Furniture            97049.37790
Name: profit, dtype: float64
```

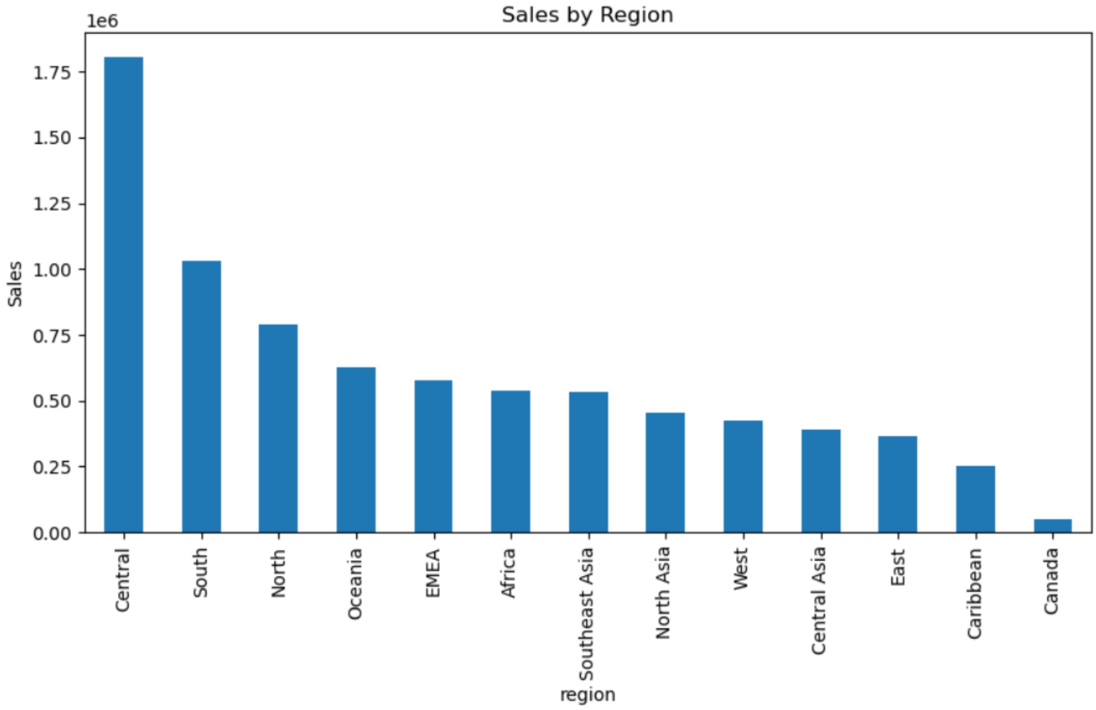


3. Sales by Region

16]:

```
region_sales = df.groupby('region')['sales'].sum().sort_values(ascending=False)

print(region_sales)

plt.figure(figsize=(10,5))
region_sales.plot(kind='bar')

plt.title("Sales by Region")
plt.ylabel("Sales")
plt.show()
```

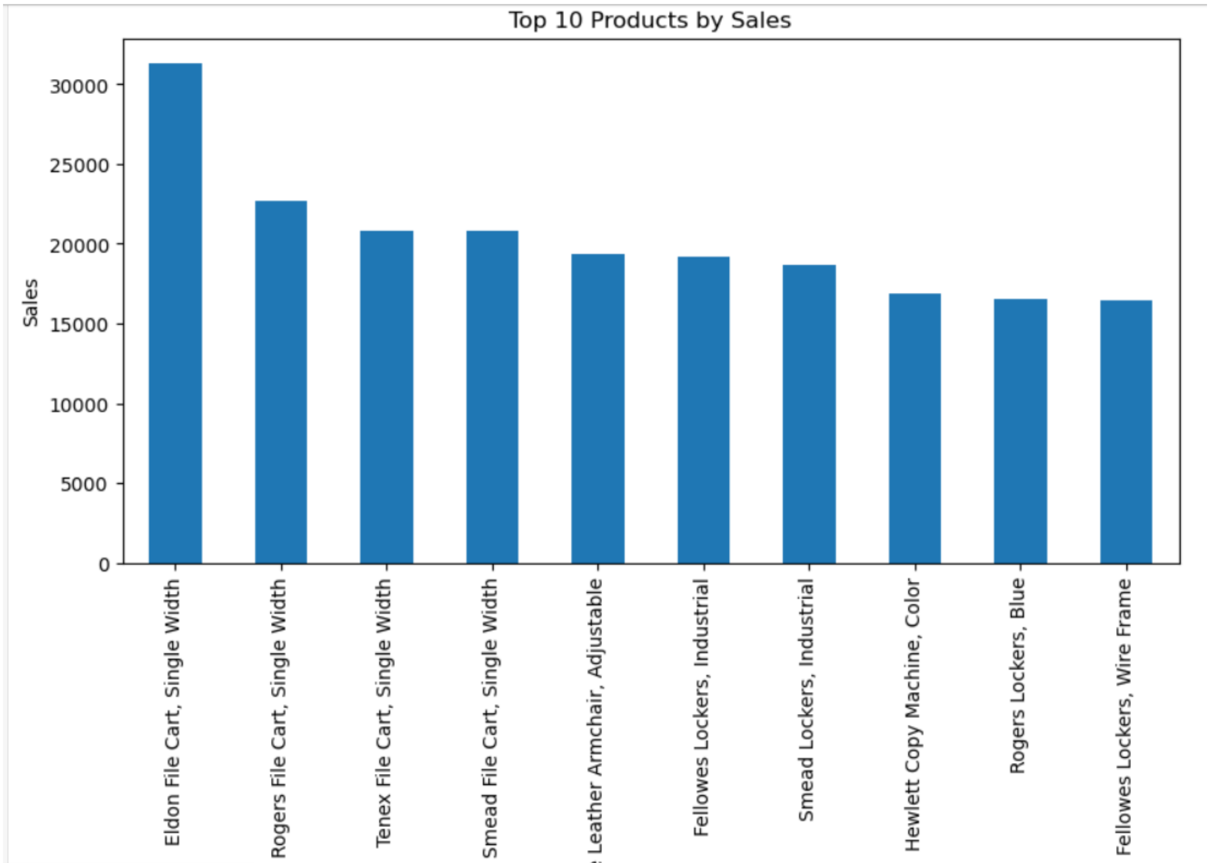
```
region
Central           1806638.0
South             1031101.0
North              790546.0
Oceania            625382.0
EMEA               575562.0
Africa             538115.0
Southeast Asia     532172.0
North Asia         453686.0
West               424173.0
Central Asia       389506.0
East               366492.0
Caribbean          251441.0
Canada              50314.0
Name: sales, dtype: float64
```



4. Top 10 Products by Sales

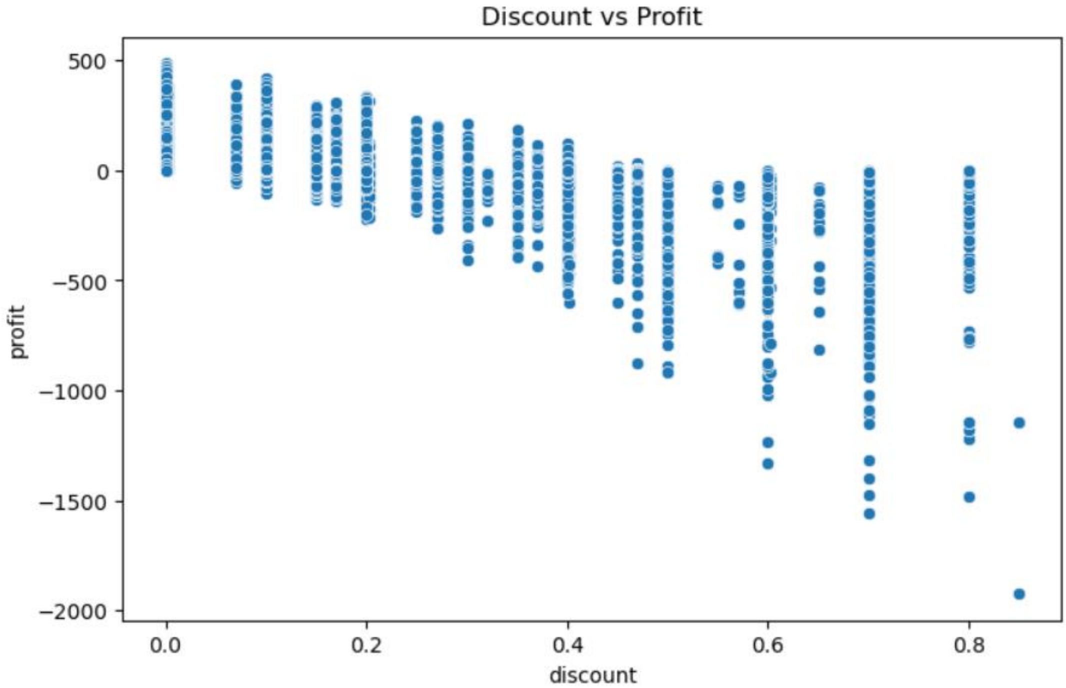
```
top_products = df.groupby('product_name')['sales'].sum().sort_values(ascending=False).head(10)

print(top_products)

plt.figure(figsize=(10,5))
top_products.plot(kind='bar')

plt.title("Top 10 Products by Sales")
plt.ylabel("Sales")
plt.show()
```

```
product_name
Eldon File Cart, Single Width                          31319.0
Rogers File Cart, Single Width                         22645.0
Tenex File Cart, Single Width                          20778.0
Smead File Cart, Single Width                          20775.0
Office Star Executive Leather Armchair, Adjustable     19355.0
Fellowes Lockers, Industrial                           19172.0
Smead Lockers, Industrial                              18648.0
Hewlett Copy Machine, Color                            16849.0
Rogers Lockers, Blue                                   16494.0
Fellowes Lockers, Wire Frame                           16470.0
Name: sales, dtype: float64
```



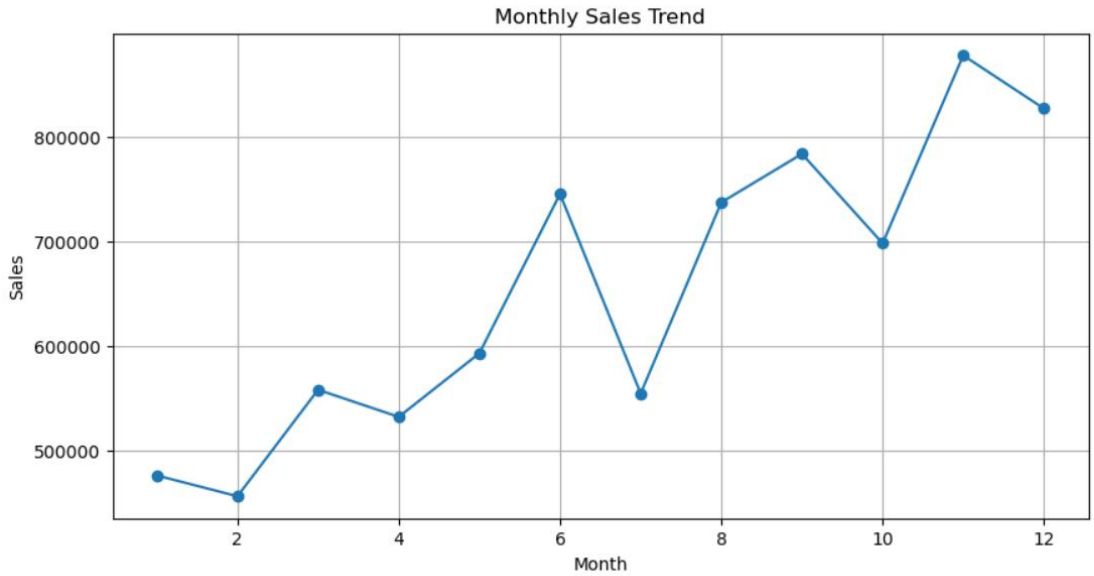
5. Discount vs Profit Analysis

3]:
```
plt.figure(figsize=(8,5))

sns.scatterplot(
    data=df,
    x='discount',
    y='profit'
)

plt.title("Discount vs Profit")
plt.show()
```



6. Monthly Sales Trend

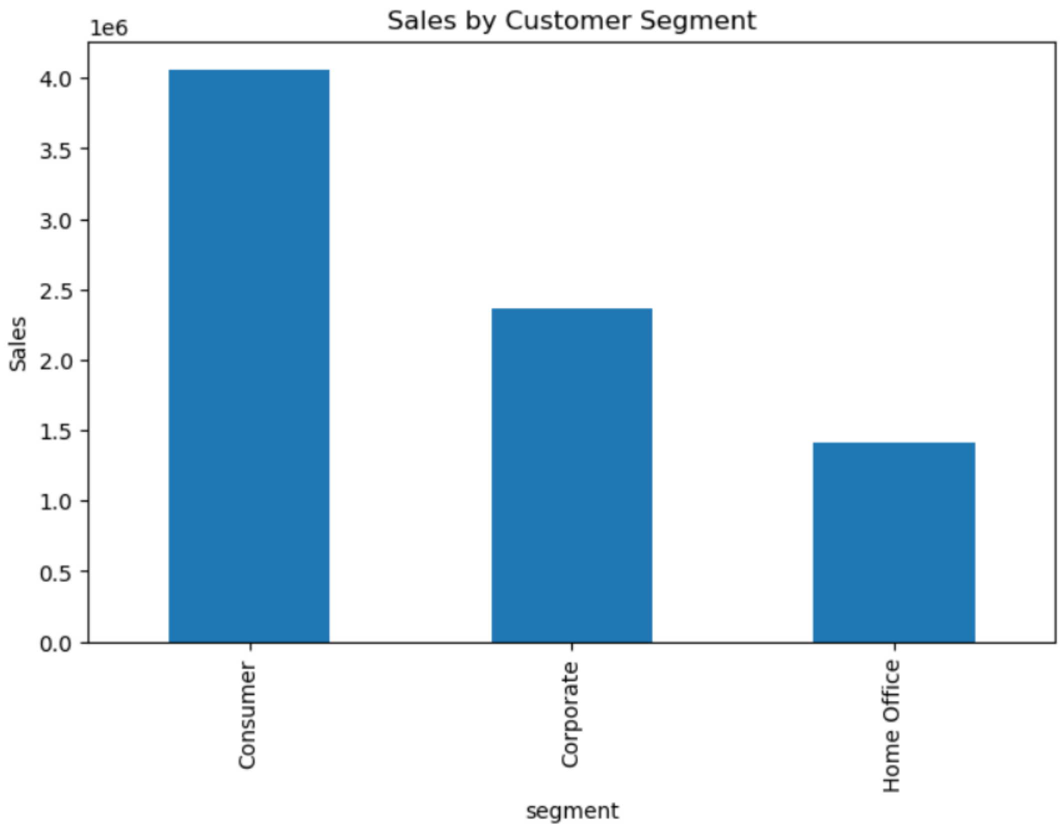
19]:
```
df['month'] = df['order_date'].dt.month

monthly_sales = df.groupby('month')['sales'].sum()

plt.figure(figsize=(10,5))

monthly_sales.plot(marker='o')

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.grid(True)

plt.show()
```



7. Customer Segment Analysis

```
]:  segment_sales = df.groupby('segment')['sales'].sum()

    print(segment_sales)

    plt.figure(figsize=(8,5))

    segment_sales.plot(kind='bar')

    plt.title("Sales by Customer Segment")
    plt.ylabel("Sales")

    plt.show()
```

```
segment
Consumer       4058118.0
Corporate      2369261.0
Home Office    1407749.0
Name: sales, dtype: float64
```



GITHUB LINK :- https://github.com/TejasSinha2006/Sales-Performance-Analysis